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Rules Contributing to Suspicious AWS IAM Activity Alert

The following rules are used to identify suspicious activity within AWS IAM logs. Any one or more of these will trigger a Suspicious AWS IAM Activity Alert. Details for each rule can be viewed by clicking the **More Details** link in the description.

Title	Description

AWS User Login Profile Was Modified	An attacker with the iam:UpdateLoginProfile permission on other users can change the password used to login to the AWS console on any user that already has a login profile setup. With this alert, it is used to detect anyone is changing password on behalf of other users.
	
	More details
	
	Rule ID
	
	`aws_2`
	
	Query
	
	`{'selection_source': {'eventSource': 'iam.amazonaws.com', 'eventName': 'UpdateLoginProfile'}, 'filter': {'userIdentity_arn|contains': 'requestParameters.userName'}, 'condition': 'selection_source and not filter'}`
	
	Log Source
	
	Stellar Cyber AWS configured for:
	
	• AWS Cloudtrail
	
	Rule Source
	
	SigmaHQ,055fb148-60f8-462d-ad16-26926ce050f1
	
	Author: toffeebr33k
	
	Tactics, Techniques, and Procedures
	
	TA0003, T1098
	
	References
	
	• https://github.com/RhinoSecurityLabs/AWS-IAM-Privilege-Escalation
	
	Severity
	
	75

Title	Description
	Suppression Logic Based On • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: test **Creation Date**: 2021/08/09 **Risk Level**: high **False Positives**: • Legit User Account Administration

AWS IAM Backdoor Users Keys	Detects AWS API key creation for a user by another user. Backdoored users can be used to obtain persistence in the AWS environment. Also with this alert, you can detect a flow of AWS keys in your org.
	
	More details
	
	Rule ID
	`aws_13`
	
	Query
	`{'selection_source': {'eventSource': 'iam.amazonaws.com', 'eventName': 'CreateAccessKey'}, 'filter': {'userIdentity_arn|contains': 'responseElements.accessKey.userName'}, 'condition': 'selection_source and not filter'}`
	
	Log Source
	Stellar Cyber AWS configured for:
	• AWS Cloudtrail
	
	Rule Source
	SigmaHQ,0a5177f4-6ca9-44c2-aacf-d3f3d8b6e4d2
	
	Author: faloker
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	• https://github.com/RhinoSecurityLabs/pacu/blob/866376cd711666c775bbfcde0524c817f2c5b181/pacu/modules/iam__backdoor_users_keys/main.py
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity: test
	Creation Date: 2020/02/12
	Risk Level: medium
	False Positives:
	• Adding user keys to their own accounts (the filter cannot cover all possible variants of user naming)
	• AWS API keys legitimate exchange workflows

Title	Description

AWS IAM User Addition to Group	Identifies the addition of a user to a specified group in AWS Identity and Access Management (IAM).
	
	More details
	
	Rule ID
	`aws_24`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'AddUserToGroup'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	N/A
	
	Severity
	25
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity: production
	Creation Date: 2020/06/04
	Risk Level: low
	False Positives
	• Adding users to a specified group may be done by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. User additions from unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS IAM Group Creation	Identifies the creation of a group in AWS Identity and Access Management (IAM). Groups specify permissions for multiple users. Any user in a group automatically has the permissions that are assigned to the group.
	
	More details
	
	Rule ID
	`aws_26`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateGroup'}, 'condition': 'selection1 and selection2'}`
	
	Log Source

Title	Description
	Stellar  AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0003, T1136 **References** N/A **Severity** 25 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2020/06/05 **Risk Level**: low **False Positives**: • A group may be created by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Group creations by unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS IAM Assume Role Policy Update	Identifies attempts to modify an AWS IAM Assume Role Policy. An adversary may attempt to modify the AssumeRolePolicy of a misconfigured role in order to gain the privileges of that role.
	
	More details
	
	Rule ID
	
	`aws_42`
	
	Query
	
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'UpdateAssumeRolePolicy'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	
	Stellar Cyber AWS configured.
	
	Rule Source
	
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	
	TA0004, T1078
	
	References
	N/A
	
	Severity
	
	25
	
	Suppression Logic Based On
	• service_id

Title	Description
	• eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2020/07/06 **Risk Level**: low **False Positives**: • Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Policy updates from unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS IAM Deactivation of MFA Device	Identifies the deactivation of a specified multi-factor authentication (MFA) device and removes it from association with the user name for which it was originally enabled. In AWS Identity and Access Management (IAM), a device must be deactivated before it can be deleted.
	
	More details
	
	Rule ID
	`aws_44`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': ['DeactivateMFADevice', 'DeleteVirtualMFADevice']}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0040, T1531
	
	References
	N/A
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	
	Maturity: production
	Creation Date: 2020/05/26
	Risk Level: medium
	False Positives:
	• A MFA device may be deactivated by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. MFA device deactivations from unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

Title	Description

AWS IAM Group Deletion	Identifies the deletion of a specified AWS Identity and Access Management (IAM) resource group. Deleting a resource group does not delete resources that are members of the group; it only deletes the group structure.
	
	More details
	
	Rule ID
	`aws_50`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'DeleteGroup'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0040, T1531
	
	References
	N/A
	
	Severity
	25
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity: production
	Creation Date: 2020/05/21
	Risk Level: low
	False Positives
	• A resource group may be deleted by a system administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Resource group deletions by unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS New MFA Method Registered For User	The following analytic identifies the registration of a new Multi Factor authentication method for an AWS account. Adversaries who have obtained unauthorized access to an AWS account may register a new MFA method to maintain persistence.
	
	More details
	
	Rule ID
	`aws_75`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateVirtualMFADevice'}, 'condition': 'selection1 and selection2'}`

Title	Description
	Log Source Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0006, T1556 **References** • https://aws.amazon.com/blogs/security/you-can-now-assign-multiple-mfa-devices-in-iam/ • https://twitter.com/jhencinski/status/1618660062352007174 **Severity** 80 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2023-01-31 **Risk Level**: medium **False Positives** • Newly onboarded users who are registering an MFA method for the first time will also trigger this detection.

AWS IAM User Created	A new account has been created in AWS IAM.
	More details
	Rule ID
	`aws_80`
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateUser'}, 'condition': 'selection1 and selection2'}`
	Log Source
	Stellar Cyber AWS configured.
	Rule Source
	Developed internally by Stellar Cyber
	Tactics, Techniques, and Procedures
	TA0003, T1136
	References
	N/A
	Severity
	50
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName

Title	Description
	• errorCode • stellar.rule_id **Additional Information** **Maturity**: test **Creation Date**: 2022/01/05 **Risk Level**: medium **False Positives**: N/A

Created AWS IAM Credentials	New IAM credentials have been generated.
	
	More details
	
	Rule ID
	`aws_81`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateAccessKey'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	N/A
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity: test
	Creation Date: 2022/01/05
	Risk Level: medium
	False Positives: N/A

IAM Policy Modification	The IAM policies associated with a user have been modified.
	
	More details
	
	Rule ID
	`aws_82`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'UpdateUserAccessPolicy'}, 'selection3': {'eventName': 'DeleteUserAccessPolicy'}, 'selection4': {'eventName': 'AddAccessPolicyToGroup'}, 'selection5': {'eventName': 'AddUserToGroup'}, 'selection6': {'eventName': 'RemoveUsersFromGroup'}, 'condition': 'selection1 and (selection2 or selection3 or selection4 or selection5 or selection6)'}`

Title	Description
	Log So Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0003, T1098 **References** N/A **Severity** 50 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: test; **Creation Date**: 2022/01/05; **Risk Level**: medium; **False Positives**: N/A

AWS IAM AccessDenied Discovery Event	The following detection identifies AccessDenied event. It is possible that an access key to AWS may have been stolen and is being misused to perform discovery events.
	More details
	Rule ID
	`aws_83`
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'errorCode': 'AccessDenied'}, 'selection3': {'userIdentity_type': 'IAMUser'}, 'selection4': {'userAgent': '*.amazonaws.com'}, 'condition': 'selection1 and selection2 and selection3 and not selection4'}`
	Log Source
	Stellar Cyber AWS configured.
	Rule Source
	Developed internally by Stellar Cyber
	Tactics, Techniques, and Procedures
	TA0007, T1580
	References
	• https://aws.amazon.com/premiumsupport/knowledge-center/troubleshoot-iam-permission-errors/
	Severity
	20
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName

Title	Description
	• errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2021-11-12 **Risk Level**: medium **False Positives**: • It is possible to start this detection will need to be tuned by source IP or user.

AWS IAM Delete Policy	The following detection identifies when a policy is deleted on AWS. This does not identify whether successful or failed, but the error messages tell a story of suspicious attempts.
	
	More details
	
	Rule ID
	
	`aws_84`
	
	Query
	
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'DeletePolicy'}, 'selection3': {'userAgent': '*.amazonaws.com'}, 'condition': 'selection1 and selection2 and not selection3'}`
	
	Log Source
	
	Stellar Cyber AWS configured.
	
	Rule Source
	
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	
	TA0003, T1098
	
	References
	
	• https://docs.aws.amazon.com/IAM/latest/APIReference/API_DeletePolicy.html
	• https://docs.aws.amazon.com/cli/latest/reference/iam/delete-policy.html
	
	Severity
	
	20
	
	Suppression Logic Based On
	
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	
	Maturity: production
	Creation Date: 2021-04-01
	Risk Level: medium
	False Positives:
	• This detection will require tuning to provide high fidelity detection capabilities. Tune based on src addresses (corporate offices, VPN terminations) or by groups of users. Not every user with AWS access should have permission to delete policies (least privilege). In addition, this may be saved separately and tuned for failed or success attempts only.

AWS IAM Failure Group Deletion	This detection identifies failure attempts to delete groups. We want to identify when a group is attempting to be deleted, but either access is denied, there is a conflict or there is no group. This is indicative of administrators performing an action, but also could be suspicious behavior occurring.

Title	Description

	More details
	
	Rule ID
	`aws_85`
	
	Query
	`{'selection2': {'eventSource': 'iam.amazonaws.com'}, 'selection3': {'eventName': 'DeleteGroup'}, 'selection4': {'errorCode': ['NoSuchEntityException', 'DeleteConflictException']}, 'selection5': {'errorCode': 'AccessDenied'}, 'selection6': {'userAgent': '*.amazonaws.com'}, 'condition': 'selection2 and selection3 and (selection4 or selection5) and not selection6'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	• https://awscli.amazonaws.com/v2/documentation/api/latest/reference/iam/delete-group.html
	• https://docs.aws.amazon.com/IAM/latest/APIReference/API_DeleteGroup.html
	
	Severity
	10
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity: production
	Creation Date: 2021-04-01
	Risk Level: medium
	False Positives:
	• This detection will require tuning to provide high fidelity detection capabilities. Tune based on src addresses (corporate offices, VPN terminations) or by groups of users. Not every user with AWS access should have permission to delete groups (least privilege).

AWS SetDefaultPolicyVersion	This search looks for AWS CloudTrail events where a user has set a default policy versions. Attackers have been know to use this technique for Privilege Escalation in case the previous versions of the policy had permissions to access more resources than the current version of the policy
	
	More details
	
	Rule ID
	`aws_86`
	
	Query
	`{'selection2': {'eventName': 'SetDefaultPolicyVersion'}, 'selection3': {'eventSource': 'iam.amazonaws.com'}, 'condition': 's`

Title	Description
	elect[redacted]2 and selection3'} **Log Source** Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0005, T1078.004 **References** • https://bishopfox.com/blog/privilege-escalation-in-aws • https://rhinosecuritylabs.com/aws/aws-privilege-escalation-methods-mitigation-part-2/ **Severity** 50 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2021-03-02 **Risk Level**: medium **False Positives**: • While this search has no known false positives, it is possible that an AWS admin has legitimately set a default policy to allow a user to access all resources. That said, AWS strongly advises against granting full control to all AWS resources
AWS Create Policy Version to allow all resources	This search looks for AWS CloudTrail events where a user created a policy version that allows them to access any resource in their account. More details **Rule ID** aws_87 **Query** {'selection2': {'eventName': 'CreatePolicyVersion'}, 'selection3': {'eventSource': 'iam.amazonaws.com'}, 'selection4': {'errorCode': 'success'}, 'condition': 'selection2 and selection3 and selection4'} **Log Source** Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0005, T1078.004 **References** • https://bishopfox.com/blog/privilege-escalation-in-aws

Title	Description
	• https://rhinosecuritylabs.com/aws/aws-privilege-escalation-methods-mitigation-part-2/ **Severity** 70 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • aws.errorCode • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2022-05-17 **Risk Level**: medium **False Positives**: • While this search has no known false positives, it is possible that an AWS admin has legitimately created a policy to allow a user to access all resources. That said, AWS strongly advises against granting full control to all AWS resources and you must verify this activity.

AWS CreateLoginProfile	This search looks for AWS CloudTrail events where a user A (victim A) creates a login profile.
	
	More details
	
	Rule ID
	
	`aws_119`
	
	Query
	
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateLoginProfile'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	
	Stellar Cyber AWS configured.
	
	Rule Source
	
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	
	TA0003, T1136.003
	
	References
	• https://bishopfox.com/blog/privilege-escalation-in-aws
	• https://rhinosecuritylabs.com/aws/aws-privilege-escalation-methods-mitigation-part-2/
	
	Severity
	
	90
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information

Title	Description
	Maturity / **Creation Date** / **Risk Level** / **False Positives** production / 2021-07-19 / medium • While this search has no known false positives, it is possible that an AWS admin has legitimately created a login profile.

AWS CreateAccessKey	This search looks for AWS CloudTrail events where a user creates access keys.
	
	More details
	
	Rule ID
	`aws_120`
	
	Query
	`{'selection1': {'eventSource': 'iam.amazonaws.com'}, 'selection2': {'eventName': 'CreateAccessKey'}, 'selection3': {'userAgent': 'console.amazonaws.com'}, 'selection4': {'errorCode': 'success'}, 'condition': 'selection1 and (selection2 and (not selection3) and selection4)'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1136.003
	
	References
	• https://bishopfox.com/blog/privilege-escalation-in-aws
	• https://rhinosecuritylabs.com/aws/aws-privilege-escalation-methods-mitigation-part-2/
	
	Severity
	70
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• aws.errorCode
	• stellar.rule_id
	
	Additional Information
	Maturity / **Creation Date** / **Risk Level** / **False Positives**
	production / 2022-03-03 / medium
	• Unknown

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